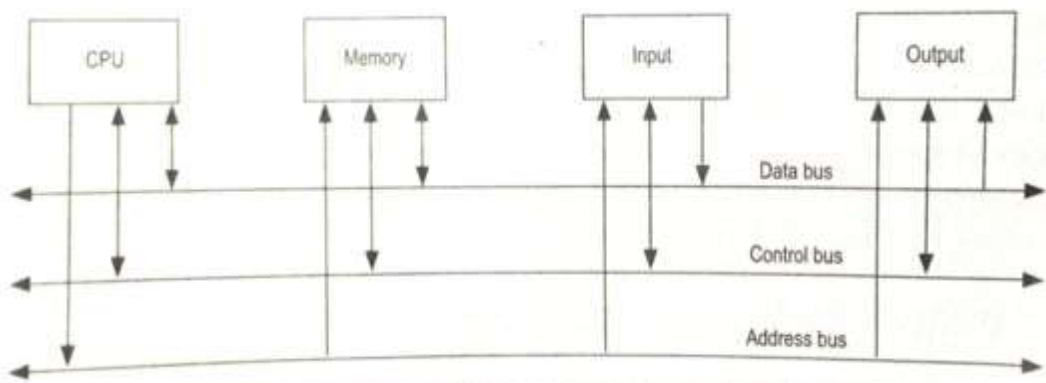


Unit – 2 Fundamental of 8085 Microprocessor

Introduction: - A *microprocessor* is a programmable electronics chip that has computing and decision making capabilities similar to central processing unit of a computer. Any microprocessor-based systems having limited number of resources are called *microcomputers*. Nowadays, microprocessor can be seen in almost all types of electronics devices like mobile phones, printers, washing machines etc. Microprocessors are also used in advanced applications like radars, satellites and flights. Due to the rapid advancements in electronic industry and large scale integration of devices results in a significant cost reduction and increase application of microprocessors and their derivatives.

Basic System Bus Architecture: The following microprocessor based computer system is as shown below fig.



The system consist of different blocks like CPU (microprocessor) , Buses Memory, Input Output units.

1. Microprocessor (CPU) :- A typical microprocessor consists of arithmetic and logic unit (ALU) in association with control unit to process the instruction execution. Almost all the microprocessors are based on the principle of store-program concept. In *store-program concept*, programs or instructions are sequentially stored in the memory locations that are to be executed.

Each microprocessor has a set of instructions. The instruction set of a microprocessor is provided in two forms: *binary machine code* and *mnemonics*. Microprocessor communicates and operates in binary numbers 0 and 1.

2. Memory Word: The number of bits that can be stored in a register or memory element is called a memory word.

3. Bus: A bus is a group of wires/lines that carry similar information.

System Bus: The system bus is a group of wires/lines used for communication between the microprocessor and peripherals (devices).

Address Bus: It carries the address, which is a binary pattern used to identify a memory location or an I/O port. The 8085 microprocessor has 16 bit address bus which carries 64K different addresses (0000 to FFFF).

Data Bus: The data bus is used to transfer data between memory and processor or between I/O device and processor. The 8085 microprocessor has 8 bit data bus, means it can transfer 8 bit data from data bus.

Control Bus: The control bus carry control signals, which consists of signals for selection of memory or I/O device from the given address, direction of data transfer and synchronization of data transfer in case of slow devices.

4. Input and OUT:- The Input devices which are used to give information to microprocessor which we have to process(execute).

Outputs are the device which accepts the processed information from the processor.

Classification of Microprocessors: Based on their specification, application and architecture microprocessors are classified.

Based on size of data bus:

- ☐ 4-bit microprocessor
- ☐ 8-bit microprocessor
- ☐ 16-bit microprocessor
- ☐ 32-bit microprocessor

Based on application:

- ☐ General-purpose microprocessor- used in general computer system and can be used by programmer for any application. Examples, 8085 to Intel Pentium.
- ☐ Microcontroller- microprocessor with built-in memory and ports and can be programmed for any generic control application. Example, 8051.
- ☐ Special-purpose processors- designed to handle special functions required for an application. Examples, digital signal processors and application-specific integrated circuit (ASIC) chips.

Concepts of T-state (Tri state devices):- Tri state logic device have three states – logic 0, logic – 1 and high impedance. The term tri state is a trade mark of National semiconductor and is used to represent three logic states. A tri state logic device has third line called as Enable. When this line is activated the tri state device function is like a ordinary device.

When the third line is disabled the logic device goes in the high impedance state as if it were disconnected from the system. Generally current is required to drive the device in logic 0 and logic 1 states. In high impedance state practically no current by the system.

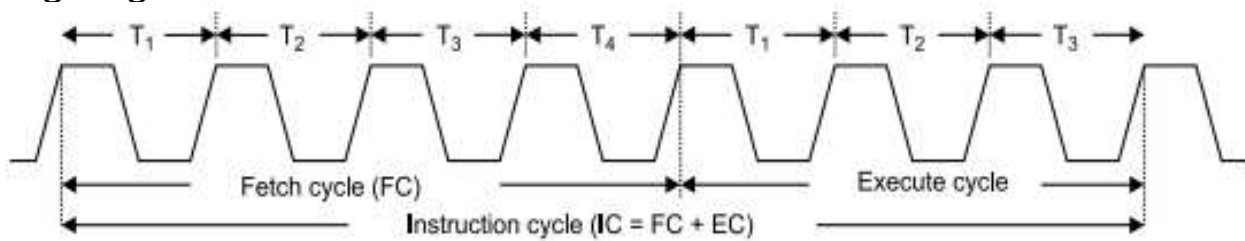
Symbols for the tri state are as shown below. There are of two tri state inverters known as active high and active low.

In microcomputer system peripherals are connected in parallel between the address bus and data bus. In the tri state interfacing devices peripherals do not load the system bus. The microprocessor communicates with one device at a time by enable ling the tri state line of the interfacing devices. The tri state logic is critical to proper functioning of the microcomputer.

Machine cycle, Instruction cycle:

1. Instruction cycle: This term is defined as the number of steps required by the CPU to complete the entire process i.e. fetching (compile) and execution of one instruction. The fetch and execute cycles are carried out in synchronization with the clock.

Timing Diagram:



Where, Instruction cycle= Fetch Cycle(FC) +Decode + Execute cycle(EC).

2. Machine cycle: It is the time required by the microprocessor to complete the operation of accessing the memory devices or I/O devices. In machine cycle various operations like opcode fetch, memory read, memory write, I/O read, I/O write are performed.

Rules to identify number of machine cycles in an instruction:

1. If an addressing mode is direct, immediate or implicit then No. of machine cycles = No. of bytes.
2. If the addressing mode is indirect then No. of machine cycles = No. of bytes + 1. Add +1 to the No. of machine cycles if it is memory read/write operation.
3. If the operand is 8-bit or 16-bit address then, No. of machine cycles = No. of bytes +1.
4. These rules are applicable to 80% of the instructions of 8085.

The machine cycles (timing diagram) for to opcode fetch (instruction), memory read memory write IO read and IO write operation are different

8085 MICROPROCESSOR

The 8085 microprocessor is an 8-bit processor available as a 40-pin IC package and uses +5 V for power. It can run at a maximum frequency of 3 MHz. Its data bus width is 8-bit and address bus width is 16-bit, thus it can address $2^{16} = 64$ KB of memory. This microprocessor can store 64KB data.

Pin Diagram of 8085:- 8085 up is an 8-bit general purpose microprocessor capable of addressing 64Kb of memory.

The device has 40 pins in DIP and work on, +5 V power supply, operate on 3 to 5 MHz frequency single phase clock. The pin function for 8085 microprocessor is as shown below.

All the signals can be classified in 8085 up pin diagram into six groups –

1) Address Bus: in this 16 signals lines. These lines are splits into two segments -

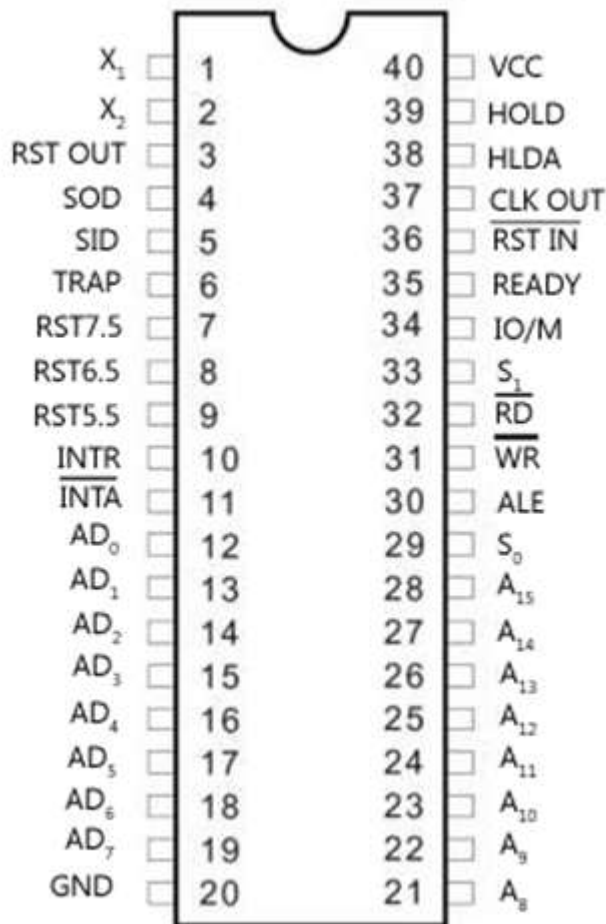
- a) $A_{15}-A_8$ – unidirectional and used for the higher order address (MSB).
- b) AD_7-AD_0 – Dual purpose such as data bus as well as lower address data bus(LSB).

2) Control and status signals: These signals are used to identify the nature of operation. **Three control signals that are-**

RD – it is a active low signal. Which indicate that the selected IO or Memory device is to be read and data is available on the data bus.

WR-it is a active low signal which indicate that the data on the data bus are to be written into a selected memory or IO location.

ALE- it is a +ve going pulse generated everytime the 8085 begins an operation (machine cycle): which indicate that the bits on AD_7-AD_0 are address bits.



Three status signals that are –

IO/M- this is a status signal used to differentiate between IO and Memory operations. When it is high then IO operation and when it is low then Memory operation.

S1 and S0- status signals, similar to IO/M, can identify various operations that are rarely used in the systems.

3) Power supply: VCC: +5 V VSS: Ground

4) Clock Frequency: X1, X2: A crystal (RC, LC N/W) is connected at these two pins. This frequency is internally divided by 2.

CLK OUT: clock output this signal can be used as the system clock for the other devices.

5) Externally initiated signals: In this

Five interrupt signals: TRAP, RST 7.5, RST 6.5, RST 5.5, and INTR (Interrupt Request)

INTA: interrupt acknowledge

RESET IN: It is an active low signal. When active program counter is set to zero.

RESET OUT: This signal indicates that the MPU is being reset; the signal can be used to reset other devices.

Architecture of 8085 Microprocessor: -

The internal architecture of 8085 is as shown below. The functional block diagram or architecture of 8085 Microprocessor, gives the complete details about the **internal Microprocessor**.

It includes the ALU (Arithmetic and logic unit), timing and control unit, instruction registers and decoder, register array, interrupt control, and serial I/O control etc.

1. The ALU (Arithmetic/Logic unit):

The ALU performs the arithmetic operations and logical operations. This unit includes the accumulator, registers, arithmetic and logic circuits and five flags.

Accumulator (A):

The accumulator is an 8-bit register for storing 8-bit data to perform arithmetic and logical operations. Any process in this can be execute with respect to accumulator and after process result also can store in accumulator. It is also called register A.

Registers (General purpose Registers):

The 8085 microprocessor has six general-purpose registers to store 8-bit data during program execution. These registers are identified as B, C, D, E, H, and L. They can be combined as register pairs-BC, DE, and HL-to perform some 16-bit operations. These registers are also called programmable registers because a programmer can use them to load or transfer the data from registers by using instructions.

Also two additional registers called temporary registers W and Z to hold 8-bit temporary data internally during the program execution. These temporary registers are not available for programmer.

The temporary registers and general purpose programmable registers are commonly called as **Register Array**.

All the signals can be classified in 8085 up pin diagram into six groups –

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a) $A_{15}-A_8$ – unidirectional and used for the higher order address (MSB).

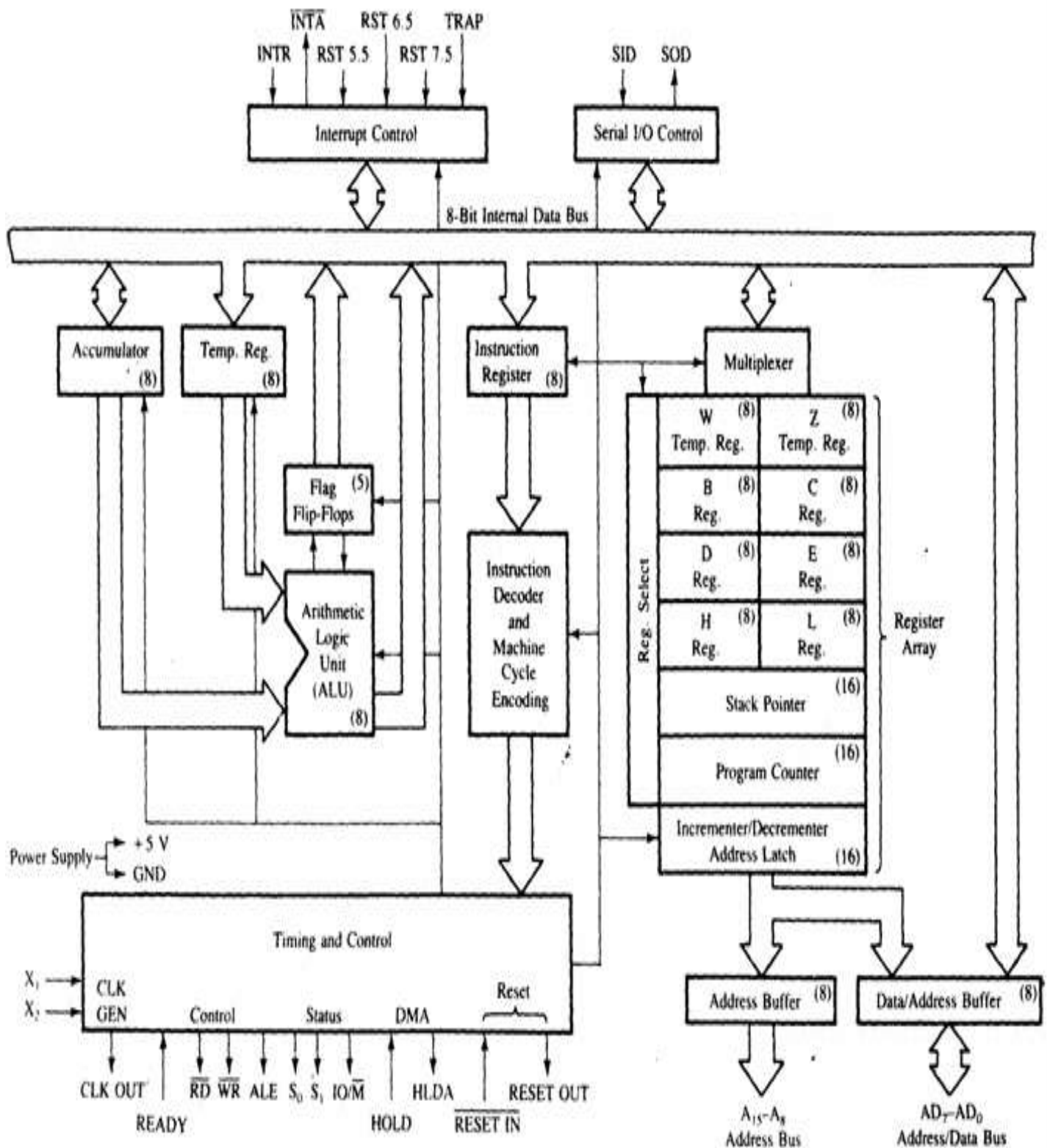
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ALE- it is a +ve going pulse generated everytime the 8085 begins an operation (machine cycle): which indicate that the bits on AD_7-AD_0 are address bits.



Flags (Flip/Flop):

The ALU having five flags which gives the status of the result stored in an accumulator and other registers. Each flag registers are one bit length. Bit position of each flag is fixed in register. The bit positions of these flags in flag registers are as follows.

D7	D6	D5	D4	D3	D2	D1	D0
SF	ZF		AF		PF		Cy

S- Sign flag: After the execution of ALU operation, if bit D7 of the result is 1 (stored in accumulator), the sign flag is set and the number will be a negative. If D7 is 0, the sign flag is reset and the number will be positive.

Z- Zero flag: The zero flag is set to 1 if the ALU operation result is zero, and the flag is set to 0, if the ALU operation result is not zero.

AC-Auxiliary carry flag: In an arithmetic operation. When a carry is generated by digit D3 and passed on to digit D4, the AC flag is set to 1.

P-Parity flag: If the result has an even number of 1's, the flag is set to 1.

If it has an odd number of 1's, the flag is reset.

I.e. 0000 0011 – even parity and 0001 0011 – odd parity

CY- Carry flag; if the results in a carry, the carry flag is set. Otherwise it is reset.

The carry flag is also set with borrow in subtractions.

Program Counter (PC):

This 16-bit register deals with sequencing the execution of instructions. This register is a memory pointer. The microprocessor uses this register to sequence the execution of the instructions. The function of the program counter is to point to the memory address from which the next byte is to be fetched. When a byte is being fetched, the program counter is automatically incremented by one to point to the next memory location.

Stack Pointer (SP):- The stack pointer is also a 16-bit register, used as a memory pointer. It points to a memory location in R/W memory, called stack. The beginning of the stack is defined by loading 16-bit address in the stack pointer. Instruction Register/Decoder It is an 8-bit register that temporarily stores the current instruction of a program. Latest instruction sent here from memory prior to execution. Decoder then takes instruction and decodes or interprets the instruction. Decoded instruction then passed to next stage.

Control and Timing Unit:- Control Unit Generates signals on data bus, address bus and control bus within microprocessor to carry out the instruction, which has been decoded.

Typical buses and their timing are described as follows:

Data Bus: Data bus carries data in binary form between microprocessor and other external units such as memory. Data bus is bidirectional. The data bus width of 8085 microprocessor is 8-bit.

Address Bus: The address bus carries addresses and is one way bus from microprocessor to the memory or other devices. 8085 microprocessor contain 16-bit address bus and are generally identified as A0 - A15.

Control Bus: Control bus is various lines which have specific functions for coordinating and controlling microprocessor operations. The control bus carries control signals partly unidirectional and partly bidirectional. The following control and status signals are used by 8085 processor:

1. ALE (output): Address Latch Enable is a pulse that is provided when an address appears on the AD0 – AD7 lines, after which it becomes 0.

2. RD (active low output): The Read signal indicates that data are being read From the selected I/O or memory device and that they are available on the data bus.

3. WR (active low output): The Write signal indicates that data on the data bus Are to be written into a selected memory or I/O location.

4. IO/M (output): It is a signal that distinguished between a memory operations And an I/O operation. When IO/M= 0 it is a memory operation and IO/M=1 it is an I/O operation.

5. S1 and S0 (output): These are status signals used to specify the type of Operation being performed which is as shown in following table.

S0	S1	Status signals and associated operations
0	0	Halt
0	1	Write
1	0	Read
1	1	Fetch

Instruction Set of 8085 Microprocessor

An instruction is a binary pattern designed inside a microprocessor to perform a specific function. Each instruction is represented by 8 bit binary value. There are following type of instruction set.

1. Data transfer Instructions
2. Arithmetic Instructions.
3. Logical Instructions.
4. Branch Instructions.
5. Machine control Instructions.

1) Data transfer instructions: In this type of Instructions, which are used to transfer data from one register to another register, from memory to register or register to memory without any change.

Examples are: MOV, MVI, LXI, LDA, STA, IN, OUT, etc.

When an instruction of data transfer group is executed, data is transferred from the source to the destination without altering the contents of the source.

e.g. when MOV A, B is executed the content of the register B is copied into the register A, and the content of register B remains unaltered.

When LDA 2500 is executed the content of the memory location 2500 is loaded into the accumulator. But the content of the memory location 2500 remains unaltered.

2) Arithmetic instructions: The instructions of this group perform arithmetic operations such as addition, subtraction; increment or decrement of the content of a register or memory.

- a. **Addition:** Any 8 bit no., the contents of a register, the content of memory location can be added to the content of Accumulator and sum (result) can be stored in the accumulator. No two other registers are added without accumulator. (i.e. ADD B,C)
e.g. Add A,B (Add content of A with content of B register and result stored in A)
- b. **Subtraction:** - Any 8 bit no., the contents of a register, the content of memory location can be subtracted from content of Accumulator and result can be stored in the accumulator. Also subtraction can be performed by using 2's compliment method. If result is -ve are expressed in 2's compliment method. No two other registers are subtracted without accumulator. (i.e. SUB B,C)
- c. **Increment/Decrement:** - The 8/ 16 bit content of a register or a memory location can be incremented or decremented by 1.

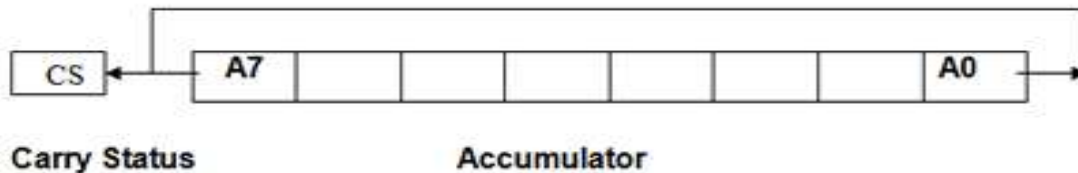
These incremented and decremented operations differ from addition and subtraction in an important way, i.e. they can be performed in any one of the registers or in memory location. After execution of the instruction, content of concern register/ memory increase/ decrease by one bit.

e.g. INR A, INR M, DCR B, DCR M.

3. Logical instructions: - The Instructions under this group perform logical operation such as AND, OR, compare, rotate etc.

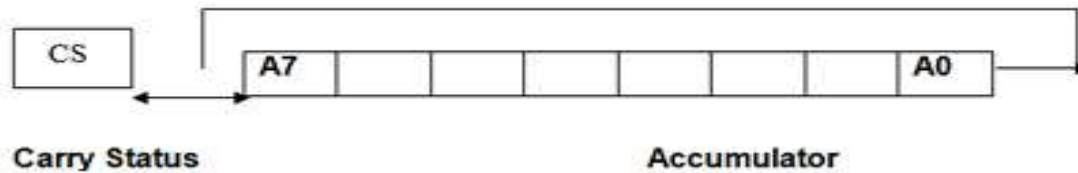
- a. **AND, OR, X OR :-** Any 8 bit no. or the content of a register, or of a memory location can be logically AND, OR, X - OR with the content of the Accumulator. After execution result can stored in accumulator.
e.g. AND A, AND M, OR A, OR M etc.
- b. **Rotate:** - Each bit in the accumulator can be shifted either left or right to the next position.
e.g. RAR, ROL.

RLC (Rotate accumulator left)



The content of the accumulator is rotated left by one bit. The seventh bit of the accumulator is moved to carry bit as well as to the zero bit of the accumulator. Only CS flag is affected.

RRC. (Rotate accumulator right)



The content of the accumulator is rotated right by one bit. The zero bit of the accumulator is moved to the seventh bit as well as to carry bit. Only CS flag is affected.

- c. **Compare:-** Any 8 bit no. or the content of a register, or of a memo location can be compare for equality , greater than, less than with the content of the accumulator.
- d. **Compliment:-** The content of accumulator can be complimented . i. e. 0 replaced by 1 and 1 replaced by 0.
e.g. CMA, CMC etc.

4. Branch Instructions:- This group includes the instructions for conditional and unconditional jump, subroutine call and return, and restart. After execution of this type of instructions program execution sequence are changed according to the instruction type.

- a. **Conditional Jump address:** After the execution of the conditional jump instruction the program jumps to the instruction specified by the address (label) if the specified condition is fulfilled. The program proceeds further in the normal sequence if the specified condition is not fulfilled. There are following type of conditional instructions.

- e.g. **JZ** addr (label). (Jump if the result is zero)
 - JNZ** addr (label) (Jump if the result is not zero)
 - JC** addr (label). (Jump if there is a carry)
 - JNC** addr (label). (Jump if there is no carry)
 - JP** addr (label). (Jump if the result is plus)
 - JM** addr (label). (Jump if the result is minus)
 - JPE** addr (label) (Jump if even parity)
 - JPO** addr (label) (Jump if odd parity)

- b. **CALL addr (label) (Unconditional CALL:** Call the subroutine identified by the operand) CALL instruction is used to call a subroutine. Before the control is transferred to the subroutine, the address of the next instruction of the main program is saved in the stack. The content of the stack pointer is decremented by two to indicate the new stack top. Then the program jumps to subroutine starting at address specified by the label.

- E.g. JMP, CALL etc.

- c. RET (Return from subroutine).
 - d. RST n (Restart) Restart is a one-word CALL instruction. The content of the program counter is saved in the stack. The program jumps to the instruction starting at restart location.
- 6. **Machine control instructions:** This type of instructions are control the system. These instructions like HALT, Interrupt. Etc.

E.g. HLT (Halt), NOP

Addressing Modes in 8085:- In different types of instructions data can transfer or copy from source to destination. The source and destination in fact are the operands. The various formats of specifying the operands are called addressing modes. There are following types of addressing modes used in 8085 microprocessor.

1. Immediate Addressing Mode
2. Register Addressing Mode
3. Direct Addressing Mode
4. Indirect Addressing Mode
5. Implied/implicit Addressing Mode

1. **Immediate Addressing:** In this type of addressing modes data are directly present in the instructions.

e.g. MVI A, 22H (since 22 data directly present in the instruction)

2. **Register Addressing:** In this type of addressing modes data can transfer from one register to another register.

e.g. MOV A, B (since data from B register transfer in accumulator(A))

3. **Direct Addressing:** In this type of addressing modes address of the data directly given in the instructions.

e.g. IN Port 04 (since port 04 address directly given in instruction)

OUT port 05

4. **Register Indirect Addressing:** In this type of addressing modes we can transfer data from memory or register. i.e. memory can used in the instructions.

e.g. MOV M,C (since moves the contents of the C register into the memory address stored in the H and L register pair.

LDAX B: - loads the accumulator with the byte of data specified by the address in the B and C register pair.

5. **Implied Addressing:** In this type of addressing mode of certain instructions is implied by the instructions function.

e.g. The STC (set carry flag) instruction deals only with the carry flag,

The DAA (decimal adjust accumulator) instruction deals with the accumulator.

Instruction format for 8085 Microprocessor:-

In 8085 microprocessor instructions are classified on the basis of word length and these are given below in detail.

1. One byte Instructions Or One Word Instructions
2. Two byte Instructions Or Two Word Instructions
3. Three byte Instructions Or Three Word Instructions

1. One Byte Instructions: This instruction uses a single Byte. That single byte specifies both opcode i.e. (Operation Code) and Operand. It consists of opcode and operand. E.g.

Instruction	Opcode	Operand	Binary Code	Hexadecimal Code
MOV D,A	MOV	D, A	01 010 011	57H(one Byte)
ADD C	ADD	C	1000 0001	81H(One Byte)
RAL	RAL	————	0001 0111	17H(One byte)

2. Two byte Instructions: In this first byte specifies the operation code and the second byte specify operand data. e.g.

Instruction	Opcode	Operand	Binary Code	Hexadecimal Code
MVI A	Data MVI	A, Data(8-bit)	0011 1110	3EH(First Byte) Data(8 bit) Second Byte

3. Three Byte Instruction: In the three byte instruction first byte specifies the opcode, second and third byte specify the (16 bit) memory address.

Instruction	Opcode	Operand	Binary Code	Hexadecimal Code
LDA 2800 H	LDA	2800 H	0011 0010 0000 0000 0010 0101	3A H (First Byte)00 H (Second Byte)28 H (Third Byte)
JMP 2050 H	JMP	2050 H	1100 0011 0101 0000 0010 0000	C3(First Byte)50 (Second Byte)20(Third Byte)

Assembly language programming (Instruction Format):- To write a ASM program using 8085 microprocessor, The commonalty used format of an instruction in ASM language is given below.

Memory Address	Machine code	Label	Mnemonics	Operands	Comments
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Memory Address:- This is the address of the memory location in which a program or a series of instructions are stored.

Machine codes:- Every instruction has one byte unique code called as operation code. Instructions are operated using data. Data may be one or two bytes. Machine codes are the hexadecimal representation of operation code.

Labels:- It is assigned for the instructions in which it appears. The presences of a label in an instruction are optional. Which a label is present, It provides a symbolic name that can be used in branch instructions of the instruction. If there is a no label then th colon must not be entered. A label may be any length of 1 to 35 characteristics. This is used for conditional or unconditional instructions.

Operands:- Operands depends on the types of instructions. In one byte instruction there is a no operands. Only one operands present in a two byte instruction and a three byte instruction has two operands, which are separated by commas.

Comment:- In this field a general comments about the instructions are always incorporated to understand the program easily. It is optional.

Assembly language programs

E.g. 1. To writes Assembly language programs for data transfer between one register to another register or in memory.

Memory Address	Machine code	Label	Mnemonics	Operands	Comments
2100		Data transfer	MVI	A, 55H	55H data immediately transfer if A register
2102			MOV	A, B	Content store in B register transfer in Accumulator.
2103			STA	2150	Transfer the content of Accumulator at the address 2150
2106			RST	5	Halt (stop the program.

To writes Assembly language programs for Arithmetic operation using 8085 microprocessor.

a. Addition

Memory Address	Machine code	Label	Mnemonics	Operands	Comments
2100		Addition	MVI	A, 55H	55H data immediately transfer if A register
2102			MVI	B, 22H	22H data immediately transfer if B register
2104			ADD	A,B	Add the content stored in A with content stored in B register, result stored in A
2105			HLT		Halt (stop) the program.

b. Subtraction.

Memory Address	Machine code	Label	Mnemonics	Operands	Comments
2100		Subtraction	MVI	A, 55H	55H data immediately transfer if A register
2102			MVI	B, 22H	22H data immediately transfer if B register
2104			SUB	A, B	Subtract the content stored in B from the content stored in B register, result stored in A
2105			HLT		Halt (stop) the program.

C. Increment

Memory Address	Machine code	Label	Mnemonics	Operands	Comments
2100		Increment	MVI	A, 55H	55H data immediately transfer if A register

2102			INR	A, 01H	22H data immediately transfer if B register
2104			HLT	5	Halt (stop) the program.

e. Decrement

Memory Address	Machine code	Label	Mnemonics	Operands	Comments
2100		Decrement	MVI	A, 55H	55H data immediately transfer if A register
2102			DCR	A, 01	22H data immediately transfer if B register
2104			HLT		Halt (stop) the program.

To write a ASM programs for Logical operations if 8085 Microprocessor.

1. AND

Memory Address	Machine code	Label	Mnemonics	Operands	Comments
2100		AND Logic	MVI	A, 55H	55H data immediately transfer if A register
2102			MVI	B, 22H	22H data immediately transfer if B register
2104			ANA	A, B	ANDed the content stored in B to the content stored in B register, result stored in A
2105			HLT		Halt (stop) the program.

2. OR

Memory Address	Machine code	Label	Mnemonics	Operands	Comments
2100		OR Logic	MVI	A, 55H	55H data immediately transfer if A register
2102			MVI	B, 22H	22H data immediately transfer if B register
2104			OR	A, B	ORed the content stored in B to the content stored in B register, result stored in A
2105			HLT		Halt (stop) the program.

Branch Operations: - To write a ASM program for addition of 05 nos.

Memory Address	Machine code	Label	Mnemonics	Operands	Comments
2400		Addition Of 10 no.	MVI	A, 00H	00H data immediately transfer if A register
2402			MVI	C, 05H	Set C register as a counter(05) transfer in C register
2404	LOOP		ADD	A, C	ADD the content stored in B to the content stored in C register, result stored in A
2405			DCR	C	Decrement content of C register By one
2406			JNZ	2404	If content of C register are 0 then go to next, otherwise JMP to loop
2409			HLT		Halt (stop) the program.

Instruction Set of 8085 Microprocessor Examples**1 . Data Transfer Group:**

The data transfer instructions move data between registers or between memory and registers.

MOV	Move
MVI	Move Immediate
LDA	Load Accumulator Directly from Memory
STA	Store Accumulator Directly in Memory
LHLD	Load H & L Registers Directly from Memory
SHLD	Store H & L Registers Directly in Memory
An 'X' in the name of a data transfer instruction implies that it deals with a register pair (16-bits);	
LXI	Load Register Pair with Immediate data
LDAX	Load Accumulator from Address in Register Pair
STAX	Store Accumulator in Address in Register Pair
XCHG	Exchange H & L with D & E
XTHL	Exchange Top of Stack with H & L

2 . Arithmetic Group:

The arithmetic instructions add, subtract, increment, or decrement data in registers or memory.

ADD	Add to Accumulator
ADI	Add Immediate Data to Accumulator
ADC	Add to Accumulator Using Carry Flag
ACI	Add Immediate data to Accumulator Using Carry
SUB	Subtract from Accumulator
SUI	Subtract Immediate Data from Accumulator
SBB	Subtract from Accumulator Using Borrow (Carry) Flag
SBI	Subtract Immediate from Accumulator Using Borrow (Carry) Flag
INR	Increment Specified Byte by One
DCR	Decrement Specified Byte by One
INX	Increment Register Pair by One
DCX	Decrement Register Pair by One

DAD Double Register Add; Add Content of Register
Pair to H & L Register Pair

3 . Logical Group:

This group performs logical (Boolean) operations on data in registers and memory and on condition flags.

The logical AND, OR, and Exclusive OR instructions enable you to set specific bits in the accumulator ON or OFF.

ANA Logical AND with Accumulator
ANI Logical AND with Accumulator Using Immediate Data
ORA Logical OR with Accumulator
OR Logical OR with Accumulator Using Immediate Data
XRA Exclusive Logical OR with Accumulator
XRI Exclusive OR Using Immediate Data

Compare Instructions:-

The Compare instructions compare the content of an 8-bit value with the contents of the accumulator;

CMP Compare
CPI Compare Using Immediate Data

Rotate Instructions

The rotate instructions shift the contents of the accumulator one bit position to the left or right:

RLC Rotate Accumulator Left
RRC Rotate Accumulator Right
RAL Rotate Left Through Carry
RAR Rotate Right Through Carry

Complement and carry flag instructions:

CMA Complement Accumulator
CMC Complement Carry Flag
STC Set Carry Flag

4 . Branch Group:

The branching instructions alter normal sequential program flow, either unconditionally or conditionally. The unconditional branching instructions are as follows:

JMP Jump
CALL Call
RET Return

Conditional branching instructions examine the status of one of four condition flags to determine whether the specified branch is to be executed. The conditions that may be specified are as follows:

NZ Not Zero (Z = 0)
Z Zero (Z = 1)
NC No Carry (C = 0)
C Carry (C = 1)
PO Parity Odd (P = 0)
PE Parity Even (P = 1)
P Plus (S = 0)
M Minus (S = 1)

Thus, the conditional branching instructions are specified as follows:

Jumps	Calls	Returns
C	CC	RC (Carry)
INC	CNC	RNC (No Carry)
JZ	CZ	RZ (Zero)
JNZ	CNZ	RNZ (Not Zero)
JP	CP	RP (Plus)

JM	CM	RM	(Minus)
JPE	CPE	RPE	(Parity Even)
JPO	CPO	RPO	(Parity Odd)

Two other instructions can affect a branch by replacing the contents of the program counter:

PCHL	Move H & L to Program Counter
RST	Special Restart Instruction Used with Interrupts

Stack I/O, and Machine Control Instructions:

The following instructions affect the Stack and/or Stack Pointer:

PUSH	Push Two bytes of Data onto the Stack
POP	Pop Two Bytes of Data off the Stack
XTHL	Exchange Top of Stack with H & L
SPHL	Move content of H & L to Stack Pointer

The I/O instructions are as follows:

IN	Initiate Input Operation
OUT	Initiate Output Operation

The Machine Control instructions are as follows:

EI	Enable Interrupt System		
DI	Disable Interrupt System		
HLT	Halt	NOP	No Operation

8085 Microprocessor Features

8085 is pronounced as "eighty-eighty-five" microprocessor. It is an 8-bit microprocessor designed by Intel in 1977 using NMOS technology.

It has the following configuration –

- 8-bit data bus
- 16-bit address bus, which can address upto 64KB
- A 16-bit program counter
- A 16-bit stack pointer
- Six 8-bit registers arranged in pairs: BC, DE, HL
- Requires +5V supply to operate at 3.2 MHZ single phase clock